Final sdn project :

Terminal 1 :

```
rakshitha@rakshitha-VirtualBox:~/pox$ cd ~/pox
./pox.py log.level --DEBUG misc.firewall
POX 0.7.0 (gar) / Copyright 2011-2020 James McCauley, et al.
INFO:misc.firewall:Firewall Controller Started
DEBUG:core:POX 0.7.0 (gar) going up...
DEBUG:core:Running on CPython (3.10.12/Mar 3 2026 11:56:32)
DEBUG:core:Platform is Linux-6.8.0-107-generic-x86_64-with-glibc2.35
WARNING:version:POX requires one of the following versions of Python: 3.6 3.7 3.8 3.9
WARNING:version:You're running Python 3.10.
WARNING:version:If you run into problems, try using a supported version.
INFO:core:POX 0.7.0 (gar) is up.
DEBUG:openflow.of_01:Listening on 0.0.0.0:6633
INFO:openflow.of_01:[00-00-00-00-00-01 2] connected
INFO:misc.firewall:Switch connected!
INFO:misc.firewall:Packet received from port 3
INFO:misc.firewall:BLOCKED h3 traffic
INFO:misc.firewall:Packet received from port 1
INFO:misc.firewall:Packet received from port 1
INFO:misc.firewall:Packet received from port 3
INFO:misc.firewall:BLOCKED h3 traffic
INFO:misc.firewall:Packet received from port 2
INFO:misc.firewall:Packet received from port 2
INFO:misc.firewall:Packet received from port 3
INFO:misc.firewall:BLOCKED h3 traffic
INFO:misc.firewall:Packet received from port 1
INFO:misc.firewall:Packet received from port 2
INFO:misc.firewall:Packet received from port 1
```

```
INFO:misc.firewall:Packet received from port 3
INFO:misc.firewall:BLOCKED h3 traffic
INFO:misc.firewall:Packet received from port 3
INFO:misc.firewall:BLOCKED h3 traffic
INFO:misc.firewall:Packet received from port 3
INFO:misc.firewall:BLOCKED h3 traffic
INFO:misc.firewall:Packet received from port 1
INFO:misc.firewall:Packet received from port 2
INFO:misc.firewall:Packet received from port 3
INFO:misc.firewall:BLOCKED h3 traffic
INFO:misc.firewall:Packet received from port 3
INFO:misc.firewall:BLOCKED h3 traffic
INFO:misc.firewall:Packet received from port 3
INFO:misc.firewall:BLOCKED h3 traffic
INFO:misc.firewall:Packet received from port 3
INFO:misc.firewall:BLOCKED h3 traffic
\INFO:misc.firewall:Packet received from port 3
INFO:misc.firewall:BLOCKED h3 traffic
```

Terminal 2

```
rakshitha@rakshitha-VirtualBox:~/pox$ sudo mn --topo single,3 \
--controller=remote,ip=127.0.0.1,port=6633 \
--mac
[sudo] password for rakshitha:
*** Creating network
*** Adding controller
*** Adding hosts:
h1 h2 h3
*** Adding switches:
s1
*** Adding links:
(h1, s1) (h2, s1) (h3, s1)
*** Configuring hosts
h1 h2 h3
*** Starting controller
c0
*** Starting 1 switches
s1 ...
*** Starting CLI:
mininet> h1 ping h2
PING 10.0.0.2 (10.0.0.2) 56(84) bytes of data.
64 bytes from 10.0.0.2: icmp_seq=5 ttl=64 time=0.194 ms
64 bytes from 10.0.0.2: icmp_seq=6 ttl=64 time=0.065 ms
64 bytes from 10.0.0.2: icmp_seq=7 ttl=64 time=0.061 ms
64 bytes from 10.0.0.2: icmp_seq=8 ttl=64 time=0.100 ms
64 bytes from 10.0.0.2: icmp_seq=9 ttl=64 time=0.064 ms
64 bytes from 10.0.0.2: icmp_seq=10 ttl=64 time=0.065 ms
64 bytes from 10.0.0.2: icmp_seq=11 ttl=64 time=0.061 ms
```

```
64 bytes from 10.0.0.2: icmp_seq=11 ttl=64 time=0.061 ms
^C
--- 10.0.0.2 ping statistics ---
11 packets transmitted, 7 received, 36.3636% packet loss, time 10287ms
rtt min/avg/max/mdev = 0.061/0.087/0.194/0.045 ms
mininet> h3 ping h2
PING 10.0.0.2 (10.0.0.2) 56(84) bytes of data.
From 10.0.0.3 icmp_seq=1 Destination Host Unreachable
From 10.0.0.3 icmp_seq=2 Destination Host Unreachable
From 10.0.0.3 icmp_seq=3 Destination Host Unreachable
From 10.0.0.3 icmp_seq=4 Destination Host Unreachable
From 10.0.0.3 icmp_seq=5 Destination Host Unreachable
From 10.0.0.3 icmp_seq=6 Destination Host Unreachable
From 10.0.0.3 icmp_seq=7 Destination Host Unreachable
From 10.0.0.3 icmp_seq=8 Destination Host Unreachable
From 10.0.0.3 icmp_seq=9 Destination Host Unreachable
^C
--- 10.0.0.2 ping statistics ---
10 packets transmitted, 0 received, +9 errors, 100% packet loss, time 9212ms
pipe 4
```

Iperf

```
mininet> h1 iperf -c h2
------------------------------------------------------------
Client connecting to 10.0.0.2, TCP port 5001
TCP window size:  400 KByte (default)
------------------------------------------------------------
[  1] local 10.0.0.1 port 54872 connected with 10.0.0.2 port 5001
[ ID] Interval       Transfer     Bandwidth
[  1] 0.0000-10.0125 sec  12.4 GBytes  10.6 Gbits/sec
mininet> h2 iperf -s &
[  1] local 10.0.0.2 port 5001 connected with 10.0.0.1 port 54872
[ ID] Interval       Transfer     Bandwidth
[  1] 0.0000-10.0087 sec  12.4 GBytes  10.6 Gbits/sec
mininet> h3 iperf -c h2
tcp connect failed: No route to host
```

Topology

```
mininet> net
h1 h1-eth0:s1-eth1
h2 h2-eth0:s1-eth2
h3 h3-eth0:s1-eth3
s1 lo:  s1-eth1:h1-eth0 s1-eth2:h2-eth0 s1-eth3:h3-eth0
c0
mininet>
```

Terminal 3

```
rakshitha@rakshitha-VirtualBox:~/pox$ sudo ovs-ofctl -O OpenFlow10 dump-flows s1
[sudo] password for rakshitha:
 cookie=0x0, duration=74.707s, table=0, n_packets=2, n_bytes=180, ipv6,vlan_tci=0x0000,dl_src=00:00:00:00:00:01,dl_dst=33:33:00:00:00:16
 actions=FLOOD
 cookie=0x0, duration=74.676s, table=0, n_packets=0, n_bytes=0, ipv6,vlan_tci=0x0000,dl_src=00:00:00:00:00:01,dl_dst=33:33:ff:00:00:01 a
ctions=FLOOD
 cookie=0x0, duration=74.427s, table=0, n_packets=0, n_bytes=0, ipv6,vlan_tci=0x0000,dl_src=00:00:00:00:00:02,dl_dst=33:33:ff:00:00:02 a
ctions=FLOOD
 cookie=0x0, duration=74.107s, table=0, n_packets=2, n_bytes=180, ipv6,vlan_tci=0x0000,dl_src=00:00:00:00:00:02,dl_dst=33:33:00:00:00:16
 actions=FLOOD
 cookie=0x0, duration=73.658s, table=0, n_packets=4, n_bytes=280, ipv6,vlan_tci=0x0000,dl_src=00:00:00:00:00:01,dl_dst=33:33:00:00:00:02
 actions=FLOOD
 cookie=0x0, duration=73.403s, table=0, n_packets=4, n_bytes=280, ipv6,vlan_tci=0x0000,dl_src=00:00:00:00:00:02,dl_dst=33:33:00:00:00:02
 actions=FLOOD
 cookie=0x0, duration=16.633s, table=0, n_packets=0, n_bytes=0, idle_timeout=30, hard_timeout=60, ipv6,vlan_tci=0x0000,dl_src=00:00:00:0
0:00:03,dl_dst=33:33:00:00:00:02 actions=drop
 cookie=0x0, duration=67.611s, table=0, n_packets=2, n_bytes=84, arp,vlan_tci=0x0000,dl_src=00:00:00:00:00:01,dl_dst=ff:ff:ff:ff:ff:ff,a
rp_spa=10.0.0.1,arp_tpa=10.0.0.2,arp_op=1 actions=FLOOD
 cookie=0x0, duration=66.556s, table=0, n_packets=1, n_bytes=42, arp,vlan_tci=0x0000,dl_src=00:00:00:00:00:02,dl_dst=00:00:00:00:00:01,a
rp_spa=10.0.0.2,arp_tpa=10.0.0.1,arp_op=2 actions=FLOOD
 cookie=0x0, duration=59.131s, table=0, n_packets=2, n_bytes=84, arp,vlan_tci=0x0000,dl_src=00:00:00:00:00:02,dl_dst=00:00:00:00:00:01,a
rp_spa=10.0.0.2,arp_tpa=10.0.0.1,arp_op=1 actions=FLOOD
 cookie=0x0, duration=58.108s, table=0, n_packets=1, n_bytes=42, arp,vlan_tci=0x0000,dl_src=00:00:00:00:00:01,dl_dst=00:00:00:00:00:02,a
rp_spa=10.0.0.1,arp_tpa=10.0.0.2,arp_op=2 actions=FLOOD
 cookie=0x0, duration=65.528s, table=0, n_packets=9, n_bytes=882, icmp,vlan_tci=0x0000,dl_src=00:00:00:00:00:01,dl_dst=00:00:00:00:00:02
,nw_src=10.0.0.1,nw_dst=10.0.0.2,nw_tos=0,icmp_type=8,icmp_code=0 actions=FLOOD
 cookie=0x0, duration=64.508s, table=0, n_packets=7, n_bytes=686, icmp,vlan_tci=0x0000,dl_src=00:00:00:00:00:02,dl_dst=00:00:00:00:00:01
,nw_src=10.0.0.2,nw_dst=10.0.0.1,nw_tos=0,icmp_type=0,icmp_code=0 actions=FLOOD
```